Denis Sergeev

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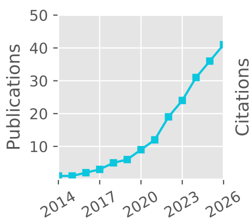

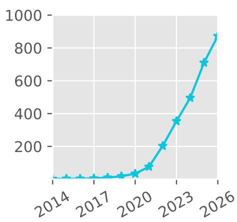

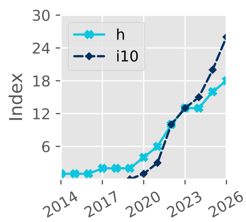

Total Pub.	**41**
Refereed	**39**
First Author	**8**
Citations	**873**
h-index	**18**

Updated: 5 Aug 2026

Career history

Jan 2025–now	**Lecturer in Astrophysics** School of Physics, University of Bristol
Sep 2021–Dec 2024	**Postdoctoral Researcher** Project: Exascale Exoplanet Modelling Department of Physics & Astronomy, University of Exeter
Sep 2018–Aug 2021	**Postdoctoral Researcher** Project: Climate Modelling of Rocky Exoplanets Department of Mathematics & Statistics, University of Exeter

Academic Qualifications

Oct 2014–Aug 2018	**PhD in Meteorology** Thesis title (shortened): 'Characteristics of Polar Lows in the Nordic Seas' School of Environmental Sciences, University of East Anglia Supervisors: Ian A. Renfrew, Thomas Spengler, Stephen Dorling
Sep 2009–May 2014	**Specialist Diploma (With Honours)** Thesis title: 'Idealised Numerical Modelling of Polar Mesocyclone Dynamics' Department of Meteorology and Climatology, Moscow State University Supervisor: Victor Stepanenko

Funding and Awards

Direct Funding, PI		**Est. Total Value**
2026	Undergraduate Student Bursary | RAS	£1500
2026	Summer Student Placement Bursary | University of Bristol	~£4500
2024	Above & Beyond Silver Award | University of Exeter	£1000
2023	Meeting Organisation Funding (Exoclimes VI and ExoSLAM) | RAS	£5000
2022	Undergraduate Student Bursary (awarded; student declined) | RAS	£1200
2017	Best Presentation Award | CEEDA Symposium	~£100
2016	Travel Bursary | Polar Prediction School	~£1000
2015	Travel Award | High-Latitude Dynamics workshop	~£1000
2014	Lord Zuckerman PhD scholarship | School of Environmental Sciences, UEA	~£112 000
2014	Young Scientist Travel Award | EGU General Assembly	~£200
2014	Russian Academy of Sciences Young Scientist Medal	~£1000
Direct Funding, co-I		
2025	Meeting Organisation Funding (UKExoM 2026) | RAS	£2500
2025	Isambard 3 Allocation (30,000 node-hours; PI: N.J. Mayne) | UKRI (Isambard 3)	▪
2024	UKSA Studentships: Mars Exploration Science	~£100 000
2024	Research Software Engineer Support | DiRAC HPC	~£45 000
Observational Facilities Resources, co-I		
2023	JWST Cycle 2, 49.21 hours (GO 3838, PI: J. Kirk)	▪

Publications

#	(preprints in **grey**)	Citations
41	Christie, D. A., Zamyatina, M., Hébrard, E., Evans-Soma, T. M., et al. (incl. **Sergeev, D. E.**), 2026, Benchmarking two chemical networks used in general circulation models of hot Jupiters, MNRAS	▪

40 Stevenson, E. T., Mak, M. T., Wolf, E., **Sergeev, D. E.**, et al., 2026, ThousandWorlds: A benchmark for climate emulation of potentially habitable exoplanets, arXiv:2606.18338 ▪

39 Nicholls, H., Krissansen-Totton, J., Lichtenberg, T., Schaefer, L., et al. (incl. **Sergeev, D.**), 2026, Coupled atmospHere Interior modeL Intercomparison (CHILI). I. Evolutionary Modelling – Primordial Magma Oceans of Earth and Venus, arXiv:2606.24757 ▪

38 Lichtenberg, T., Schaefer, L., Krissansen-Totton, J., Miguel, Y., et al. (incl. **Sergeev, D. E.**), 2026, Coupled atmospHere Interior modeL Intercomparison (CHILI)—Protocol Version 1.0: A CUISINES Intercomparison Project of Magma Ocean Models, Planet. Sci. J. **3**

37 Claringbold, A. B., Fisher, C. E., Kirk, J., Ahrer, E., et al. (incl. **Sergeev, D. E.**), 2026, BOWIE-ALIGN: Sub-solar C/O ratio and metallicity atmosphere of the misaligned hot Jupiter HAT-P-30 b, MNRAS **5**

36 Ahrer, E., Fairman, C., Kirk, J., Wakeford, H. R., et al. (incl. **Sergeev, D. E.**), 2025, BOWIE-ALIGN: weak spectral features in KELT-7b's JWST NIRSpec/G395H transmission spectrum imply a high cloud deck or a low-metallicity atmosphere, MNRAS **7**

35 Mak, M. T., **Sergeev, D. E.**, Mayne, N. J., Zamyatina, M., et al., 2025, The impact of different haze types on the atmospheres and observations of hot Jupiters: 3D simulations of HD 189733b, HD 209458b, and WASP-39b, MNRAS **3**

34 **Sergeev, D. E.**, McDermott, J. W., Woods, L., Braam, M., et al., 2025, Lightning activity on a tidally locked terrestrial exoplanet in storm-resolving simulations for a range of surface pressures, MNRAS **1**

33 Meech, A., Claringbold, A. B., Ahrer, E., Kirk, J., et al. (incl. **Sergeev, D. E.**), 2025, BOWIE-ALIGN: substellar metallicity and carbon depletion in the aligned TrES-4b with JWST NIRSpec transmission spectroscopy, MNRAS **14**

32 Kirk, J., Ahrer, E., Claringbold, A. B., Zamyatina, M., et al. (incl. **Sergeev, D. E.**), 2025, BOWIE-ALIGN: JWST reveals hints of planetesimal accretion and complex sulphur chemistry in the atmosphere of the misaligned hot Jupiter WASP-15b, MNRAS **30**

31 Penzlin, A. B. T., Booth, R. A., Kirk, J., Owen, J. E., et al. (incl. **Sergeev, D. E.**), 2024, BOWIE-ALIGN: how formation and migration histories of giant planets impact atmospheric compositions, MNRAS **43**

30 **Sergeev, D. E.**, Boutle, I. A., Lambert, F. H., Mayne, N. J., et al., 2024, The Impact of the Explicit Representation of Convection on the Climate of a Tidally Locked Planet in Global Stretched-mesh Simulations, ApJ **11**

29 Natchiar, S. R. M., Webb, M. J., Lambert, F. H., Vallis, G. K., et al. (incl. **Sergeev, D. E.**), 2024, Reduction in the Tropical High Cloud Fraction in Response to an Indirect Weakening of the Hadley Cell, JAMES **2**

28 Zamyatina, M., Christie, D. A., Hébrard, E., Mayne, N. J., et al. (incl. **Sergeev, D. E.**), 2024, Quenching-driven equatorial depletion and limb asymmetries in hot Jupiter atmospheres: WASP-96b example, MNRAS **24**

27 Mak, M. T., **Sergeev, D. E.**, Mayne, N., Banks, N., et al., 2024, 3D simulations of TRAPPIST-1e with varying CO_2, CH_4, and haze profiles, MNRAS **10**

26 Villanueva, G. L., Fauchez, T. J., Kofman, V., Alei, E., et al. (incl. **Sergeev, D. E.**), 2024, Modeling Atmospheric Lines by the Exoplanet Community (MALBEC) Version 1.0: A CUISINES Radiative Transfer Intercomparison Project, Planet. Sci. J. **13**

25 Kirk, J., Ahrer, E., Penzlin, A. B. T., Owen, J. E., et al. (incl. **Sergeev, D. E.**), 2024, BOWIE-ALIGN: A JWST comparative survey of aligned versus misaligned hot Jupiters to test the dependence of atmospheric composition on migration history, RAS Techniques and Instruments **25**

24 Mak, M. T., Mayne, N. J., **Sergeev, D. E.**, Manners, J., et al., 2023, 3D Simulations of the Archean Earth Including Photochemical Haze Profiles, J. Geophys. Res.: Atmospheres **10**

23 **Sergeev, D. E.**, Mayne, N. J., Bendall, T., Boutle, I. A., et al., 2023, Simulations of idealised 3D atmospheric flows on terrestrial planets using LFRic-Atmosphere, Geosci. Model Dev. **16**

22 Cohen, M., Bollasina, M. A., **Sergeev, D. E.**, Palmer, P. I., et al., 2023, Traveling Planetary-scale Waves Cause Cloud Variability on Tidally Locked Aquaplanets, Planet. Sci. J. **10**

21 Eager-Nash, J. K., Mayne, N. J., Nicholson, A. E., Prins, J. E., et al. (incl. **Sergeev, D. E.**), 2023, 3D Climate Simulations of the Archean Find That Methane has a Strong Cooling Effect at High Concentrations, J. Geophys. Res.: Atmospheres **9**

20 McCulloch, D., **Sergeev, D. E.**, Mayne, N., Bate, M., et al., 2023, A modern-day Mars climate in the Met Office Unified Model: dry simulations, Geosci. Model Dev. **8**

19 Braam, M., Palmer, P. I., Decin, L., Ridgway, R. J., et al. (incl. **Sergeev, D. E.**), 2022, Lightning-induced chemistry on tidally-locked Earth-like exoplanets, MNRAS **22**

18 Christie, D. A., Lee, E. K. H., Innes, H., Noti, P. A., et al. (incl. **Sergeev, D. E.**), 2022, CAMEMBERT: A Mini-Neptunes General Circulation Model Intercomparison, Protocol Version 1.0.A CUISINES Model Intercomparison Project, Planet. Sci. J. **9**

17 **Sergeev, D. E.**, Fauchez, T. J., Turbet, M., Boutle, I. A., et al., 2022, The TRAPPIST-1 Habitable Atmosphere Intercomparison (THAI). II. Moist Cases-The Two Waterworlds, Planet. Sci. J. **77**

16 Turbet, M., Fauchez, T. J., **Sergeev, D. E.**, Boutle, I. A., et al., 2022, The TRAPPIST-1 Habitable Atmosphere Intercomparison (THAI). I. Dry Cases-The Fellowship of the GCMs, Planet. Sci. J. **64**

15 Fauchez, T. J., Villanueva, G. L., **Sergeev, D. E.**, Turbet, M., et al., 2022, The TRAPPIST-1 Habitable Atmosphere Intercomparison (THAI). III. Simulated Observables-the Return of the Spectrum, Planet. Sci. J. **55**

14 **Sergeev, D. E.**, Lewis, N. T., Lambert, F. H., Mayne, N. J., et al., 2022, Bistability of the Atmospheric Circulation on TRAPPIST-1e, Planet. Sci. J. **33**

13 Cohen, M., Bollasina, M. A., Palmer, P. I., **Sergeev, D. E.**, et al., 2022, Longitudinally Asymmetric Stratospheric Oscillation on a Tidally Locked Exoplanet, ApJ **17**

12 Fauchez, T. J., Turbet, M., **Sergeev, D. E.**, Mayne, N. J., et al., 2021, TRAPPIST Habitable Atmosphere Intercomparison (THAI) Workshop Report, Planet. Sci. J. **39**

11 Terpstra, A., Renfrew, I. A., & **Sergeev, D. E.**, 2021, Characteristics of Cold-Air Outbreak Events and Associated Polar Mesoscale Cyclogenesis over the North Atlantic Region, J. Cli. **30**

10 Renfrew, I. A., Barrell, C., Elvidge, A. D., Brooke, J. K., et al. (incl. **Sergeev, D.**), 2021, An evaluation of surface meteorology and fluxes over the Iceland and Greenland Seas in ERA5 reanalysis: The impact of sea ice distribution, Q. J. R. Meteorol. Soc. **76**

9 Eager-Nash, J. K., Reichelt, D. J., Mayne, N. J., Hugo Lambert, F., et al. (incl. **Sergeev, D. E.**), 2020, Implications of different stellar spectra for the climate of tidally locked Earth-like exoplanets, A&A **28**

8 **Sergeev, D. E.**, Lambert, F. H., Mayne, N. J., Boutle, I. A., et al., 2020, Atmospheric Convection Plays a Key Role in the Climate of Tidally Locked Terrestrial Exoplanets: Insights from High-resolution Simulations, ApJ **68**

7 Joshi, M. M., Elvidge, A. D., Wordsworth, R., & **Sergeev, D.**, 2020, Earth's Polar Night Boundary Layer as an Analog for Dark Side Inversions on Synchronously Rotating Terrestrial Exoplanets, ApJ **20**

6 Renfrew, I. A., Pickart, R. S., Våge, K., Moore, G. W. K., et al. (incl. **Sergeev, D.**), 2019, The Iceland Greenland Seas Project, BAMS **28**

5 **Sergeev, D.**, Renfrew, I. A., & Spengler, T., 2018, Modification of Polar Low Development by Orography and Sea Ice, Mon. Wea. Rev. **18**

4 Shestakova, A. A., Toropov, P. A., Stepanenko, V. M., **Sergeev, D. E.**, et al., 2018, Observations and modelling of downslope windstorm in Novorossiysk, Dyn. Atm. Ocean. **7**

3 **Sergeev, D. E.**, Renfrew, I. A., Spengler, T., & Dorling, S. R., 2017, Structure of a shear-line polar low, Q. J. R. Meteorol. Soc. **22**

2 Spengler, T., Renfrew, I. A., Terpstra, A., Tjernström, M., et al. (incl. **Sergeev, D.**), 2016, High-Latitude Dynamics of Atmosphere-Ice-Ocean Interactions, BAMS **7**

1 Eliseev, A. V., & **Sergeev, D. E.**, 2014, Impact of subgrid-scale vegetation heterogeneity on the simulation of carbon-cycle characteristics, Izv. Atmos. Ocean. Phy. **9**

Conferences and Seminars

Invited Talks

May 2025 General circulation models for exoplanet atmospheres
JpGU-AGU Joint Meeting 2026 | Chiba, Japan

Oct 2025 CUISINES - a framework for exoplanet model intercomparison projects
Atmospheric and interior evolution of planetary magma oceans | Leiden, the Netherlands

Jun 2025 Atmospheric dynamics on other planets
Durham HPC Days | Durham, UK

Feb 2025 Exoplanet climate modelling with LFRic

	University of East Anglia | Norwich, UK
May 2024	3D simulations of exoplanet atmospheres with the next-generation Met Office model University of Leicester | Leicester, UK
Apr 2024	Shall I compare thee to a distant world? Inter-planet and inter-model comparative studies EGU General Assembly | Vienna, Austria
Jul 2023	Simulations of idealised 3D atmospheric flows on terrestrial planets using LFRic-Atmosphere NASA GISS Seminar | Online
Mar 2023	First results of using LFRic for exoplanet climate modelling NIWA Seminar | Wellington, New Zealand
Feb 2023	Atmospheric dynamics and chemistry on exoplanets UQ Astro Group Meeting | Brisbane, Australia
Feb 2023	Atmospheric dynamics and chemistry on exoplanets UniSQ Exoplanet Group Seminar | Brisbane, Australia
Feb 2023	Atmospheric dynamics and chemistry on exoplanets UNSW AstroSeminar | Sydney, Australia
Apr 2022	Dichotomy of the atmospheric circulation on TRAPPIST-1e NASA GISS Seminar | Online
Jan 2022	Dichotomy of the atmospheric circulation on TRAPPIST-1e NASA GSFC Extrasolar Planets Seminar | Online
Nov 2021	TRAPPIST-1 Habitable Atmosphere Intercomparison (THAI) MPIA APEx Exocoffee | Online
May 2021	Overcast on TRAPPIST-1e RCC MSU Geophysical Seminar | Online
Sep 2020	Simulations of convection over a range of atmospheric conditions on TRAPPIST-1e THAI Workshop | Online
Apr 2020	Atmospheric convection plays a key role in the climate of tidally locked exoplanets University of Reading Meteorology Seminar | Online
Apr 2020	Atmospheric convection plays a key role in the climate of tidally locked exoplanets NASA GISS Seminar | Online

Contributed Talks

May 2026	Lightning climatology on TRAPPIST-1e ELSI workshop | Tokyo, Japan
Sep 2023	Introducing GeoVista - Cartographic rendering and mesh analytics powered by PyVista (joint talk) Met Office Seminar | Exeter, UK
Jul 2022	Bistability of the atmospheric circulation on TRAPPIST-1e Rocky Worlds II | Oxford, UK
Apr 2022	Dichotomy of the atmospheric circulation on TRAPPIST-1e Exoplanet Modelling in the James Webb Era II: Terrestrial planets and sub-Neptunes | Online
Nov 2020	Explicit convection on tidally locked rocky exoplanets simulated with the UM nesting suite Unified Model users workshop | Online
Aug 2019	Simulations of moist convection on tidally-locked rocky exoplanets Exoclimes V | Oxford, UK
Jun 2019	North Atlantic polar mesoscale cyclones in ERA5 and ERA-Interim reanalyses IGP workshop | Norwich, UK
Apr 2019	Atmospheric convection on tidally-locked Earth-like exoplanets UK Exoplanet Community Meeting | London, UK
Jun 2018	Modification of Polar Low Development by Sea Ice and Svalbard Orography POLAR2018 | Davos, Switzerland
Oct 2017	The influence of Svalbard orography and sea ice on polar low development 18th Cyclone Workshop | Sainte-Adèle, Canada
Apr 2017	Polar lows and how background environment can influence their development Cambridge Earth Systems Science EnvEast Doctoral Alliance Symposium | Cambridge, UK
May 2016	Structure of the shear-line polar low south of Svalbard

	NORPAN meeting | Tokyo, Japan
Apr 2016	Structure of the shear-line polar low south of Svalbard
	13th European Polar Lows Working Group Workshop | Paris, France

Poster Presentations

Nov 2025	Lightning climatology on rocky exoplanets in a global storm-resolving model CTR Wilson Meeting on Atmospheric Electricity | Bath, UK
Jun 2024	The impact of convection on the climate of a tidally locked planet in stretched-mesh simulations Exoplanets 5 | Leiden, Netherlands
Apr 2024	The impact of convection on the climate of TRAPPIST-1e in global stretched-mesh simulations EGU General Assembly | Vienna, Austria
Apr 2024	The impact of convection on the climate of a tidally locked planet in stretched-mesh simulations UK Exoplanet Community Meeting | Birmingham, UK
Nov 2022	Dry Modern-Day Mars Climate in the Met Office Unified Model UK Solar System Planetary Atmospheres | London, UK
Sep 2022	Bistability of the Atmospheric Circulation on TRAPPIST-1e UK Exoplanet Community Meeting | Edinburgh, UK
Jul 2015	Structure and dynamics of a shear-line polar low during a cold-air outbreak over the Norwegian Sea Royal Meteorological Society Student Conference | Birmingham, UK
Mar 2015	Structure and dynamics of a shear-line polar low during a cold-air outbreak over the Norwegian Sea Dynamics of Atmosphere-Ice-Ocean Interactions in the High Latitudes workshop | Rosendal, Norway
May 2014	Numerical modelling of polar mesocyclones dynamics diagnosed by the energy budget EGU General Assembly | Vienna, Austria
Apr 2013	Impact of subgrid-scale vegetation heterogeneity on the carbon cycle EGU General Assembly | Vienna, Austria
Apr 2013	Numerical modelling of polar mesocyclones generation mechanisms EGU General Assembly | Vienna, Austria

Supervision

(Projects with me as the lead supervisor are in **bold**. Students who continued their academic career are underlined.)

PhD Supervision

Sep 2025–Sep 2029	**Alex Corbett** (U. Bristol) **Project: Convection on Sub-Neptunes** Co-supervisors: B. Shipway, Z. Leinhardt
Sep 2025–Sep 2029	Will Luscombe Project: Forecasting Martian dust storms Co-supervisors: N. J. Mayne, M. Bate, B. Drummond
Sep 2021–Apr 2025	Mei Ting (Martha) Mak (U. Exeter) Project: Hazes in Planetary Atmospheres Co-supervisors: N. J. Mayne, J. Manners, E. Hébrard

MSc Supervision

Sep 2025–May 2026	**Evie Cushing** Project: **Ice Giant Atmospheric Modelling**
Sep 2025–May 2026	**Freya Evans & Daisy Green** Project: **Atmospheric Dynamics on Ice Giants**
Sep 2025–May 2026	**Catherine Kerr & Lily Odhuba** Project: **Lightning Storms on Earth-like Exoplanets**
Jan 2023–May 2024	Tom Batchelor, Luke Benzing, & Alex McGinty Project: Mars Atmosphere Modelling Co-supervisors: M. Bate, N. J. Mayne, D. McCulloch
Sep 2020–Sep 2022	Danny McCulloch (MSci by Research)

	Project: Climate Modelling of Modern-Day Mars Co-supervisors: M. Bate, N. J. Mayne
Apr 2021–Sep 2022	Meghan Plumridge (MSci by Research) Project: Climate Modelling of Early Mars Co-supervisors: M. Bate, N. J. Mayne
Jan 2021–May 2022	Jasper Chadwick & Esse Sellwood Project: Ocean Heat Transport on Rocky Exoplanets Co-supervisors: F. H. Lambert, J. Eager-Nash
Jan 2021–May 2022	Isabelle Browne & Oakley Young Project: Greenhouse Effect on Early Mars Co-supervisors: F. H. Lambert, N. J. Mayne, J. Eager-Nash
Jan 2020–May 2021	Toby Ferrison Project: Titan Climate Modelling Co-supervisor: F. H. Lambert
Oct 2018–May 2019	Jake Eager-Nash & David Reichelt Project: Implications of Stellar Type on the Climate of Tidally Locked Terrestrial Exoplanets Co-supervisors: F. H. Lambert, N. J. Mayne

Undergraduate and Summer Internship Supervision

Jul–Sep 2022	Oakley Young Project: Ekman Ocean Model Co-supervisors: J. Eager-Nash, F. H. Lambert
Jun–Sep 2022	**James McDermott & Lottie Woods** Project: **Simulations of Lightning Storms on Tidally Locked Rocky Exoplanets**
Jun–Aug 2021	Oakley Young Project: Climate Modelling of Archean Earth Co-supervisors: J. Eager-Nash, N. J. Mayne
Jun–Aug 2021	Joshua Parkin & Esse Sellwood Project: The Impact of Host Star Spectrum on the Climate of Rocky Exoplanets Co-supervisors: J. Eager-Nash, N. J. Mayne
Jun–Aug 2019	Isobel Parry Project: Water Cycle on Proxima Centauri b Co-supervisor: F. H. Lambert

PhD Examinations

2025 Maria Di Paolo | University of East Anglia | Supervisors: Manoj Joshi, David Stevens

Teaching and Mentoring

2026–now	Environmental Physics Lecturer | University of Bristol | ∼40 students
2025–now	Practical Physics III: Research Skills and Group Project Tutor | University of Bristol | 2 groups of ∼7 students
2025–now	Research Project in Physics Supervisor & assessor | University of Bristol | ∼10 students
Jul 2024	Algorithms For Exascale Summer School Invited lecturer | University of Exeter | ∼20 students
Feb 2024	Physics of Climate Change Workshop lead | University of Exeter | ∼30 students
Jul 2023	Climatematch Academy Mentor | Online | 3 groups of ∼5 students
Jul 2023	International Sustainability Summer School Lecturer | University of Exeter | ∼10 students
Jun 2023	Exoclimes Summer School in Atmospheres and Modelling (ExoSLAM)

	Lecturer | University of Exeter | ∼50 students
2016–2018	Introduction to Python in Environmental Sciences
	Course creator & lead | University of East Anglia | ∼50 students
2015–2017	Modelling Environmental Processes; Meteorology; Numerical Skills
	Teaching assistant | University of East Anglia

Research Leadership and Impact

2024–now	Co-lead of Climates Using Interactive Suites of Intercomparisons Nested for Exoplanet Studies (CUISINES)
Jun 2023	Co-chair of Exoclimes Summer School in Atmospheres and Modelling (ExoSLAM)
2023	Interview by the University of Exeter about my research
2023	Interview by UKRI/STFC about my outreach
2023	Expert Scientist at the British Science Festival Climate Exhibition
2022	Press releases: University of Exeter, American University, & INSU CNRS
2020–now	3D visualisations of exoplanet simulations:
	'Cloudy Skies of Distant Exoplanets' | University of Exeter Images of Research 2023
	'A refined look at tidally locked exoplanets' | DiRAC HPC Research Image Competition 2023
	'Exoplanetary Atmospheres' | Exeter Science Centre, Science as Art Gallery 2020
	'Dusty exoplanet atmospheres' | Nature Press Release
	'Virtual Reality Exploration of Exoplanets' | 360 VR video (contributor)
2019	Science consulting on the 'Exoplanet Explorers' videogame
2015	Blogging:
	Disastrous Disaster Movies
	Polar Lows: What Fuels Arctic Hurricanes?
	Worldwide Weird Weather Words

Organisation of Scientific Meetings

Jun 2026	Idealised modelling with LFRic: training course (Co-Chair and Instructor) | Exeter, UK | ∼50 attendees
Mar 2026	UK Exoplanet Community Meeting (SOC) | Bristol, UK
Oct 2025	Atmospheric and interior evolution of planetary magma oceans (SOC) | Leiden, the Netherlands
Sep 2025	BUFFET-5 (Co-chair) | Bordeaux, France
Jul 2025	Exoclimes VII (SOC) | Montreal, Canada
Jun 2025	Idealised modelling with LFRic (Chair) | Exeter, UK | ∼50 attendees
Oct 2024	BUFFET-4: Building a Unified Framework For Exoplanet Treatments (Co-chair) | Online
Jun 2024	What's Cookin' Doc? A CUISINES meeting (Chair) | Leiden, the Netherlands | ∼20 attendees
Jun 2023	ExoSLAM Summer School (Co-chair) | Exeter, UK | ∼50 attendees
Jun 2023	Exoclimes VI (LOC) | Exeter, UK | ∼200 attendees
Mar 2023	Challenge of Science Leadership Short Course | Exeter, UK
Aug 2022	Exoclimatology Theory Group Summer Retreat | Mawgan Porth, UK | 15 attendees

Reviewing and Academic Service

Journals	Nat. Astron., MNRAS, Planet. Sci. J., Geophys. Res. Lett., ApJ, Planet. Space Sci., Q. J. R. Meteorol. Soc.
Funding	STFC Consolidated Grant, STFC ERF, NASA XRP, STFC Small Award
Observations	James Webb Space Telescope General Observer Programs (Exoplanets & Disks, Cycles 3 & 4)
Membership	Royal Astronomical Society, Europlanet Society

Technical Skills

Numerical models	LFRic, Unified Model, SOCRATES, LAGRANTO, Isca
Programming languages	Python, FORTRAN, MATLAB, NCL
Python libraries (user)	cartopy, cython, iris, matplotlib, numpy, pandas, pyvista, xarray

Python libraries (creator/contributor)	aeolus, cartopy, pyvista, geovista
Parallel computing	Dask, MPI, OpenMP
Version control	Git, Subversion
Document preparation	LaTeX, Quarto, Jupyter Notebooks, Markdown, HTML, CSS, reST

Vocational Training

Sep 2023	Belbin Training
Mar 2023	Challenge of Science Leadership
Dec 2022	Interview Training
Jul 2020	Writing Workshop for Climate Scientists
Mar 2020	ESA JWST Master Class
Jul 2019	ICTP Summer School on Convective Organization and Climate Sensitivity
Apr 2018	Fortran Modernisation Workshop
Jan 2018	Helicopter Underwater Escape Training Course (CA-EBS)
Dec 2017	Sea Survival Course
Jun 2017	Weather Presenting
Feb 2017	Level 1 First Aid for Field Work Course
Jan 2017	Raspberry Pi Basics
Apr 2016	WWRP/WCRP/Bolin Center Polar Prediction School
Dec 2014	UK Met Office Unified Model Training

Vocational Experience

Apr–Jun 2018	Data Technician Processing of meteorological data collected in the IGP field campaign | University of East Anglia
2015–2018	Founder and Leader Python Users Group | University of East Anglia
Feb–Mar 2018	Member of the Meteorology Team The Iceland-Greenland Seas Project (IGP) field campaign | Akureyri, Iceland
Mar 2015	Rapporteur Dynamics of Atmosphere-Ice-Ocean Interactions in the High-Latitudes | Rosendal, Norway
Oct 2013	Research Intern Geophysical Institute | University of Bergen, Norway
Aug–Sep 2013	Trainee Forecaster Forecast and Briefing Service | Main Aviation Meteorological Centre, Vnukovo Airport
Jul 2012	Research Intern A.M. Obukhov Institute of Atmospheric Physics | Moscow, Russia

Unsuccessful Funding Applications

2026	GW4 Building Communities Generator Fund
2025	STFC Small Award
2025	Leverhulme Research Leadership Award
2025	ERC Starting Grant
2025	2nd GPU Embedded CSE (eCSE) support call
2024	1st GPU Embedded CSE (eCSE) support call
2023	STFC Ernest Rutherford Fellowship
2021	Leverhulme Trust Early Career Fellowship